Evaluation Metrics

How We Know If a Model's Actually Good

You train a model, it makes predictions ...

But now what?

Evaluation metrics tell us how well a model is performing for the kind of problem we're solving.

Not All Metrics Are Created Equal

Accuracy

Simple, but not always enough.

$$\text{Accuracy} \quad \frac{\text{Correct}}{\text{Predictions}}$$

Good if your classes are balanced. Bad if you're dealing with imbalannce (e.g., 95% no, 5% yes)

Precision & Recall

Of all predicted positives, how many were correct?

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

Use Precision when false positives are moredang-

Precision & Recall

Of all predicted positives, how many were correct?

Precision =

Recall =

TP	FP
FN	TN

Use Precision when false positives are costly.

Use Recall when false negatives are more dangerous

F1 Score

Harmonic mean of precision and recall. Balances both.

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$